



Systematic Growth Factor Treatment on the Patients with Androgenetic Alopecia



Byung In Ro, Shin Han Kim, Sang Yoon Lee, Hyun Ok Son

Department of Dermatology, Myongji Hospital, Kwandong University
College of Medicine

Background

- **New emerging treatment**
 - PRP(platelet-rich plasma) injection : roller and stamp
 - Mesotherapy : roller and stamp
 - Cytokine therapy : microneedle
- YB Lee et al. Effects of topical application of growth factors followed by microneedle therapy in women with female pattern hair loss: a pilot study. J Dermatol 39, 2012
 - 5 weeks treatment with microneedle therapy
 - Half-split study / 11 patients
 - Hair increased 10% compared with baseline
- Herein, we investigated the injection of systemic growth factor for alopecia

Materials and Methods

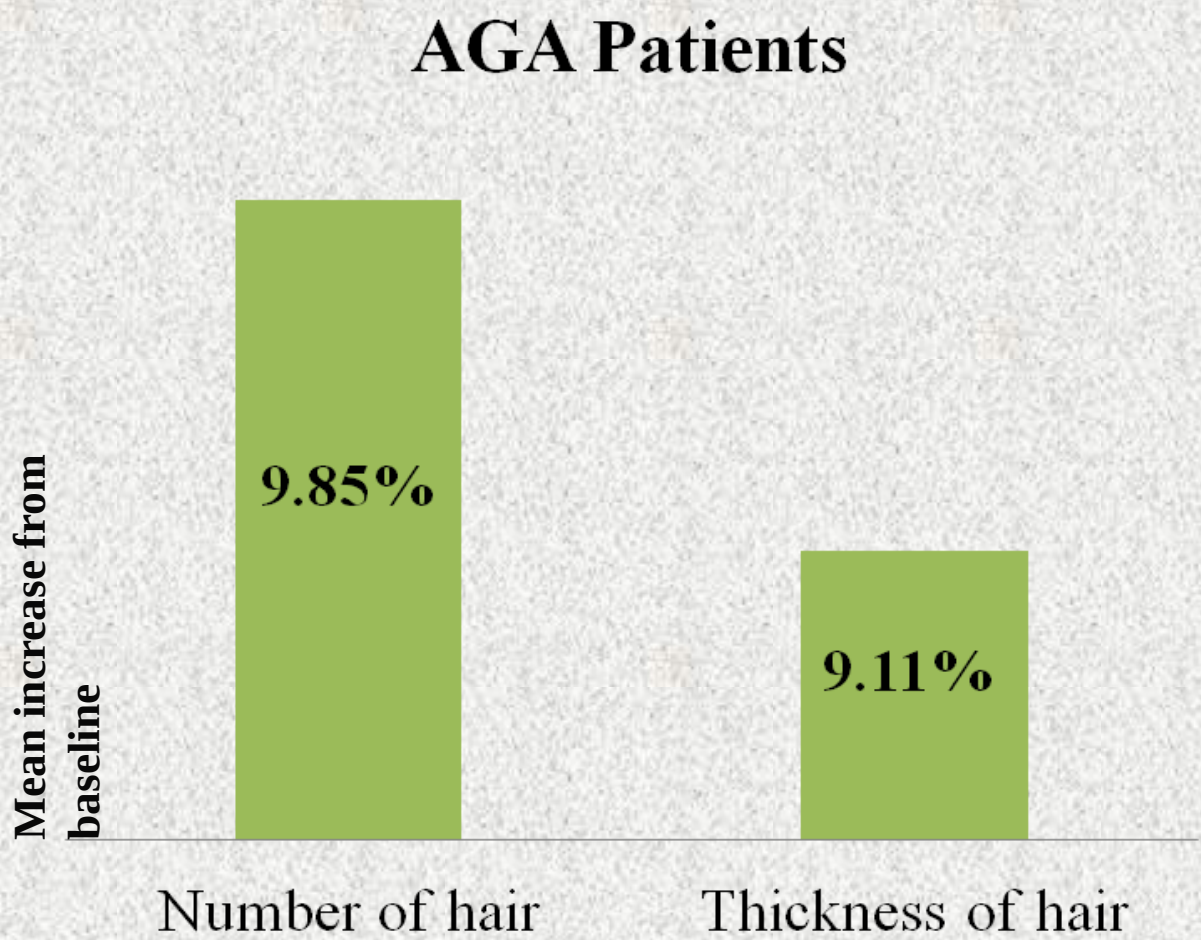
- **144 Androgenetic alopecia Patients**
 - From October 2011 to September 2012
 - 28patients excluded
 - MPHL : 54patients / FPHL : 62patients
- **Treatment**
 - Topical application of growth factor solution to scalp by microneedle therapy and electroporation
 - Growth factor (AGF39™; Aesmed Co. Ltd., Seoul, Korea)
 - FGF , IGF -1, VEGF, stem cell factor, KGF -2, superoxide dismutase-1, and Noggin
 - Microneedle (Dr Back 10 story FNS FN-1; Dongbang Medicare, Bundang, Korea)
 - Electroporation (Derma-S; Robomax, Sungnam, Korea)
- **Total 10 times (6 months) by 2~4 weeks interval**
- **Clinical photographs and phototrichogram images**
- **Efficacy: 5 grade - number and thickness of hair**
 - 0(no increase) / 1(0~5% increase) / 2(5~10% increase) / 3(10~15% increase) / 4(>15% increase)

Results

Patient’s characteristics

Androgenetic Alopecia (Total 123 patients)	Number	Mean age	Min. age	Max. age
Male	54 patients	35.4 ± 3.82	19	60
Female	62 patients	40.7 ± 4.12	19	57

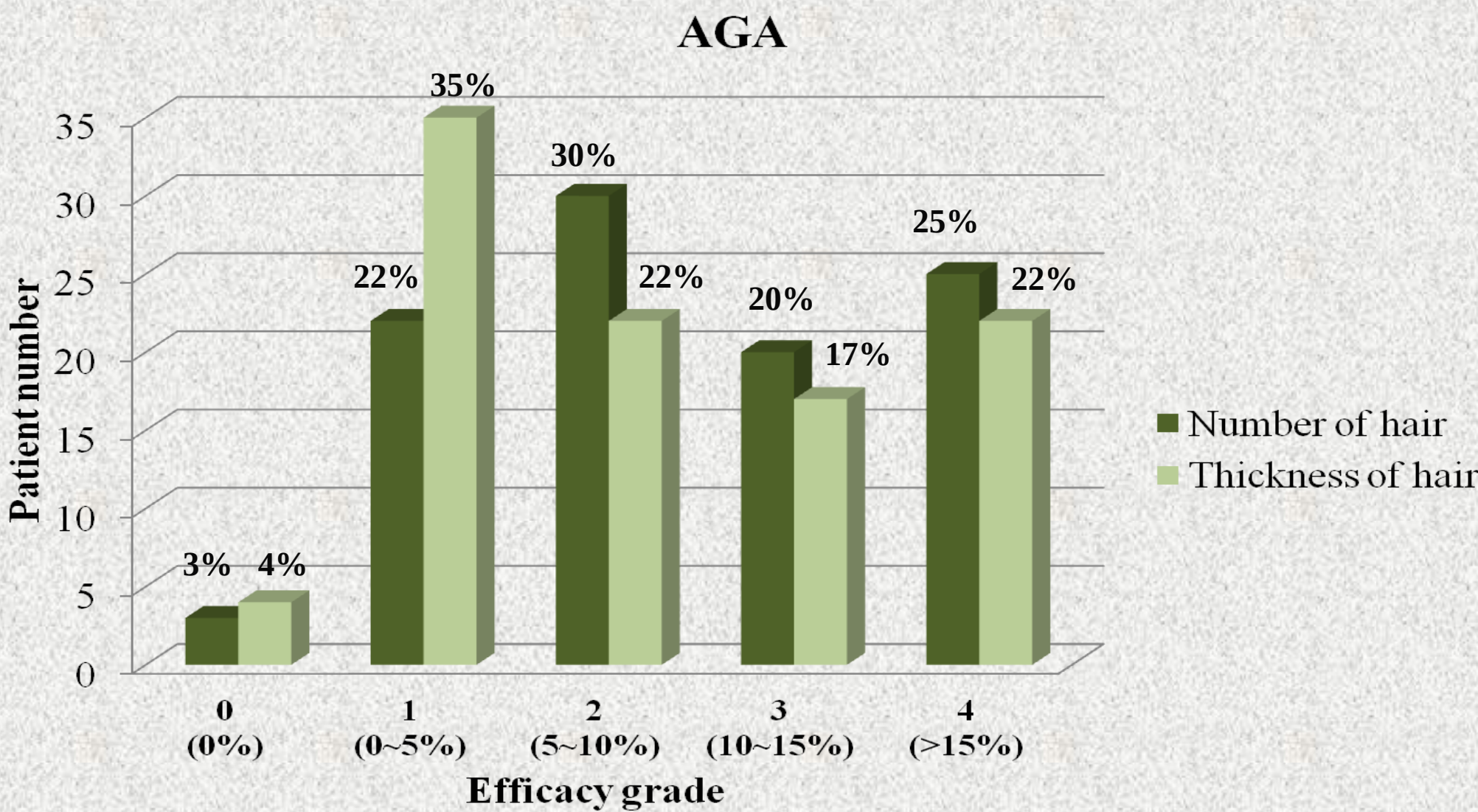
Mean increase from baseline in AGA patients



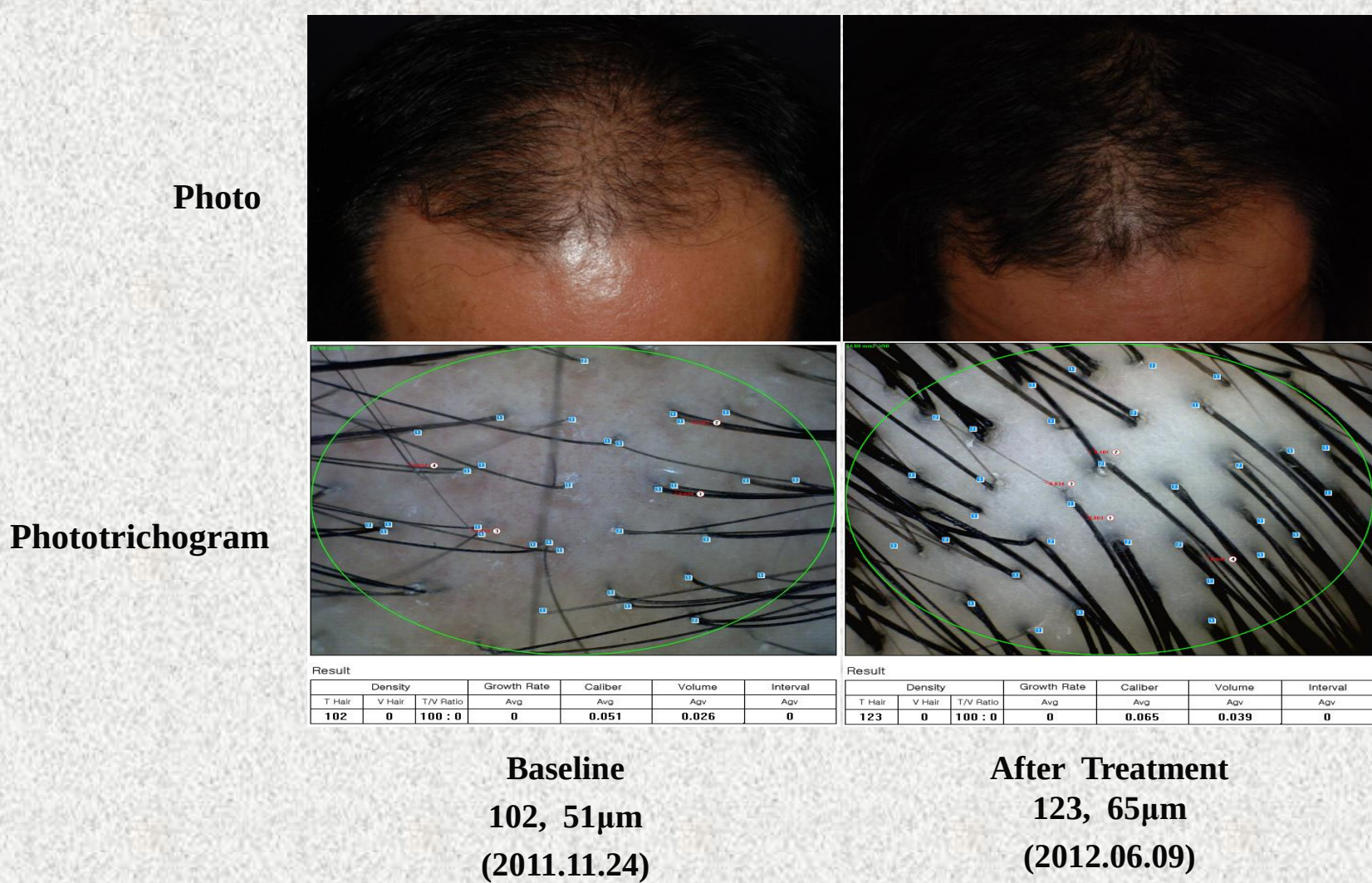
- **Increase in hair number** : 0% ~ 29.41%
- **Increase in hair thickness** : 0% ~ 27.87%

Results

Efficacy grade in AGA patients



M/47 MPHL IV



Discussion

	Lee et al	Our Study
Method	Microneedle only	Microneedle + Electroporation
Number of Patients	11	123
AGA	FPHL only	MPHL and FPHL
Follow Up Period	5 Weeks	25 Weeks
Evaluation	Digital Microscope AM313T, Dino-Lite AnMo Electronics, Taiwan	Folliscope 2.8 Lead M, Seoul, Korea

Summary

- Systemic growth factor therapy was effective and safe for the treatment of androgenetic alopecia
- Adverse effect : none, except mild tingling sensation
- This therapy will be one of the treatment options for androgenetic alopecia

References

- YB Lee et al. Effects of topical application of growth factors followed by microneedle therapy in women with female pattern hair loss: a pilot study. J Dermatol 39, 2012:21:231-4.
- Du Cros DL. Fibroblast growth factor and epidermal growth factor in hair development. J Invest Dermatol 1993;101:106S-13S.
- Braun S, Krampert M, Bodo E, Kumin A, Born-Berclaz C, Paus R et al. Keratinocyte growth factor protects epidermis and hair follicles from cell death induced by UV irradiation, chemotherapeutic or cytotoxic agents. J Cell Sci 2006;119:4841-9.